

Pre-Algebra
First Trimester Study Guide

Name _____

Period _____

1. Put the numbers in order from least to greatest. 7, 0, -1.8, -5, 5.1, -3

-5, -3, -1.8, 0, 5.1, 7

Solve each:

2. $64 + (-73)$

-9

3. $-18 - 15$

-33

4. $(5)(-2)(15)$

-150

5. $\frac{-72}{8}$

-9

Solve each using order of operations:

6. $2(4 + 3 \cdot 2) + (-3 + 5)^2$

$2(4+6) + (2)^2$
 $2(10) + 4$
 $20 + 4$
24

7. $\frac{3+7+9}{5-(-25)}$

$\frac{21+9}{5+25}$

$\frac{30}{30}$
1

8. (-5,8) is in which quadrant on the coordinate plane?

II

II	I
III	IV

Use $a = -2$ $b = 10$ and $c = -5$ to evaluate questions 9 and 10.

9. $(a + b)^2 + bc$

$(-2+10)^2 + (10)(-5)$
 $8^2 + (-50)$
 $64 + (-50)$
14

10. $\frac{abc}{2a}$

$\frac{(-2)(10)(-5)}{2(-2)}$
 $\frac{100}{-4}$
-25

Solve each:

$$11. d - 58 = 16$$

$$+58 \quad +58$$

$$d = 74$$

$$12. 12n = -48$$

$$\div 12 \quad \div 12$$

$$n = -4$$

$$13. 15n + 1 = -59$$

$$-1 \quad -1$$

$$\frac{15n}{15} = \frac{-60}{15}$$

$$n = -4$$

$$14. \frac{v}{9} - 6 = -3$$

$$+6 \quad +6$$

$$9 \cdot \frac{v}{9} = 3 \cdot 9$$

$$v = 27$$

$$15. 7l + 9 + l = -47$$

$$8l + 9 = -47$$

$$-9 \quad -9$$

$$\frac{8l}{8} = \frac{-56}{8}$$

$$l = -7$$

Write an inequality for each word expression

16. c is at most 20.

$$c \leq 20$$

17. m is more than 86

$$m > 86$$

Solve each inequality and graph the solution set on the number line provided.

$$18. f - 16 < -21$$

$$+16 \quad +16$$

$$f < -5$$

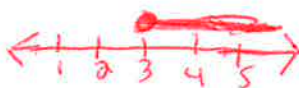


$$19. 11r + 1 \geq 34$$

$$-1 \quad -1$$

$$\frac{11r}{11} \geq \frac{33}{11}$$

$$r \geq 3$$

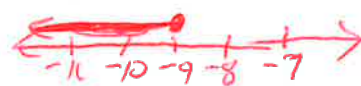


$$20. \frac{k}{9} - 3 \leq -4$$

$$+3 \quad +3$$

$$9 \cdot \frac{k}{9} \leq -1 \cdot 9$$

$$k \leq -9$$



21. Give the prime factorization of 24.

$$\begin{array}{l} 2 \times 12 \\ \quad 2 \times 6 \\ \quad \quad 2 \times 3 \\ 2^3 \times 3 \end{array}$$

22. What is the LCM of 8 and 12?

$$\begin{array}{l} 8 \\ 12 \end{array} \quad \begin{array}{l} 24 \\ 16 \\ 24 \end{array} \quad (24)$$